

SECTION A (25 marks)

This section consists of 3 questions. Answer **all** questions.

1. Evaluate each of the following limits, if exists.

a) $\lim_{x \rightarrow 3} \frac{2x-6}{-x^2+2x+3}$ [3 marks]

b) $\lim_{x \rightarrow 9} \frac{x-9}{3(\sqrt{x}-3)}$ [3 marks]

c) $\lim_{x \rightarrow -\infty} \frac{x+2}{\sqrt{4x^2+1}}$ [3 marks]

2. Differentiate the following with respect to x. Give your answer in the simplest form.

a) $y = \tan^3(e^x + 3)$ [3 marks]

b) $y = \ln\left(\frac{2-x^2}{x(1-x)}\right)$ [4 marks]

c) $y = 2^{x^3 \ln x}$ [4 marks]

3. The curve $y = 2x^3 + bx^2 + cx$ has a turning point at (1,5).

a) Find the values of b and c. [3 marks]

b) By using the second derivative test, determine whether the stationary point at (1, 5) is a minimum or maximum point. [2 marks]

SECTION B (75 marks)

This section consists of 7 questions. Answer **all** questions.

1. a) Given $x = 3 + 2i$, $y = -i$ and $\frac{z}{13} + x = \frac{1}{x} + y^2$.
Express z in polar form. [6 marks]

- b) Given that the complex number, express $z = \sqrt{-1} (2i - i^{10})$ in Cartesian form. [3 marks]

2. a) Determine the possible values of x if $\left| \frac{2x}{3+x} + 1 \right| \geq 2$.
Express your answer in interval form. [6 marks]

- b) Find the value of x such that $2|x| \geq 3x - 10$. [5 marks]

3. Given $f(x) = 2x + 1$ and $g(x) = \frac{-3x}{x+1}$.

- i) State the domain of $f(x)$ and $g(x)$. [2 marks]

- ii) Obtain $fg(x)$ and $gf(x)$. Hence, determine the constant value of k where $f \circ g(k) = g \circ f(k)$. [8 marks]

4. Given

$$f(x) = \begin{cases} -|x+2|, & -5 \leq x \leq -1 \\ 3, & -1 < x < 1 \\ (x-2)^2 + 1, & x \geq 1 \end{cases}$$

- i) Evaluate $f(-3)$. [1 marks]

- ii) Sketch the graph of $f(x)$. [3 marks]

- iii) Based on the graph, determine the point of x – intercept and y – intercept. [2 marks]

- iv) Hence, find the domain and range of f . [2 marks]

5. Given

$$f(x) = \begin{cases} ax + 7 & , \quad x < -k \\ 3 & , \quad -k \leq x < 1 \\ x^2 + 1 & , \quad 1 \leq x < k \\ 3x + 1 & , \quad k < x < 5 \\ 16 & , \quad x \geq 5 \end{cases}$$

i) Find the values of k if $\lim_{x \rightarrow k} f(x)$ exist. [5 marks]

ii) Hence, by using the value of $k > 0$, find the value of a if $\lim_{x \rightarrow -k} f(x)$ exist. [5 marks]

6. a) If $xy = 2(x - y)^2$, find the following values at $(1, \frac{1}{2})$

i) $\frac{dy}{dx}$ ii) $\frac{d^2y}{dx^2}$ [9 marks]

b) Given $x = \cos 2t$ and $y = 2t + \sin^2 t$, show that $\frac{dy}{dx} = -\operatorname{cosec} 2t - \frac{1}{2}$ and evaluate $\frac{d^2y}{dx^2}$ when $t = \frac{\pi}{8}$. [7 marks]

7. a) Given a water tank which has a shape of circular cone with a base radius of 3 metres and a height of 5 metres. If water being pumped into the tank at the rate of $2\text{m}^3/\text{minute}$, find the rate of change in water level rising when the water is 3 metres deep. [5 marks]

b) A closed cylinder of height h cm and radius r cm. The volume of the cylinder is $128\pi \text{ cm}^3$, and its total surface area is $A \text{ cm}^2$

i) Show that $A = 2\pi(r^2 + \frac{128}{r})$. [3 marks]
ii) Calculate the minimum value of A . [3 marks]

Final Answer:

SECTION A

1. a) $-\frac{1}{2}$ b) 2 c) $-\frac{1}{2}$
2. a) $3e^x \tan^3(e^x + 3) \sec^2(e^x + 3)$ b) $-\frac{2x}{2-x^2} - \frac{1}{x} + \frac{1}{1-x}$
 c) $(1+3 \ln x)x^2 2^{x^3 \ln x} \ln 2$
3. $b = -9$ $c = 12$, maximum point

SECTION B

1. a) $z = 7\sqrt{65}[\cos(-2.622) + i \sin(-2.622)]$ b) $z = -2 + i$
2. a) $(-\infty, -3) \cup \left(-3, -\frac{9}{5}\right] \cup [3, +\infty)$ b) $\{x : x \leq 10\}$
3. a) i) $D_f : (-\infty, \infty)$; $D_g : (-\infty, -1) \cup (-1, \infty)$ ii) $\frac{-6k-3}{2k+2}$, $k = \frac{5}{4}$
4. i) $f(-3) = -1$ ii) D.I.Y iii) x - intercept $(-2, 0)$, y - intercept $(0, 3)$
 iv) $D_f : [-5, \infty)$ $R_f : [-3, 0] \cup [1, \infty)$
- 5 i) $k = 0, k = 3$ ii) $a = \frac{4}{3}$
6. a) i) $\frac{1}{2}$ ii) 0 b) 2
7. a) $\frac{dh}{dt} = \frac{50}{81\pi} \text{ m/min}$ b) $A = 96\pi \text{ cm}^2$